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CURRENT OPINION

Quantification of Neuromuscular Fatigue: What Do We Do Wrong and Why?

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Abstract

Neuromuscular fatigue (NMF) is usually assessed non-invasively in healthy, athletic or clinical populations with the combination of voluntary and evoked contractions. Although it might appear relatively straightforward to magnetically or electrically stimulate at different levels (cortical/spinal/muscle) and to measure mechanical and electromyographic responses to quantify neuromuscular adjustments due to sustained/repeated muscle contractions, there are drawbacks that researchers and clinicians need to bear in mind. The aim of this opinion paper is to highlight the pitfalls inevitably faced when NMF is quantified. The first problem might arise from the definition of fatigue itself and the parameter(s) used to measure it; for instance, measuring power vs. isometric torque may lead to different conclusions. Another potential limitation is the delay between exercise termination and the evaluation of neuromuscular function; the possible underestimation of exercise-induced neural and contractile impairment and misinterpretation of fatigue etiology will be discussed, as well as solutions recently proposed to overcome this problem. Quantification of NMF can also be biased (or not feasible) because of the techniques themselves (e.g. results may depend on stimulation intensity for transcranial magnetic stimulation) or the way data are analyzed (e.g. M wave peak-to-peak vs first phase amplitude). When available, alternatives recently suggested in the literature to overcome these pitfalls are considered and recommendations about the best practices to assess NMF (e.g. paying attention to the delay between exercise and testing, adapting the method to the characteristics of the population to be tested and considering the limitations associated with the techniques) are proposed.

1 Introduction

Fatigue has been defined in numerous different ways; for example, the chronic fatigue experienced by patients compared with acute fatigue due to mental or physical exercises [1, 2]. Even when considering only acute fatigue, Angelo Mosso explained as early as in 1891 that fatigue can be defined by two series of phenomena: the diminution of

Key points

Various approaches exist to characterize neuromuscular function which can be combined to better understand human fatigability from a fundamental to an applied continuum. They all require the familiarization of the participant during a separate session, particularly if muscle/nerve/cortical stimulations are used.

Some of these techniques are uncomfortable and not suitable in frail populations. To minimize the discomfort, several techniques are presented (e.g. muscle stimulation via large electrodes, magnetic peripheral stimulation, submaximal intensities) and their limits are discussed.

Recommendations are provided for protocol design (e.g. TMS intensity, large muscle mass testing, isometric vs dynamic modes) and data analysis (e.g. separate analysis of the first and second phase of the M wave).

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muscular force and a sensation of fatigue (see recent review from Kluger et al. [3]). In the present paper, we will focus only on the first part of this definition, also known as neuromuscular fatigue (NMF) or fatigability [4].

NMF has been extensively studied for the past 100 years. There are now numerous techniques to non-invasively assess the neural and contractile determinants of impaired muscle force generating capacity. However, despite papers published on a daily basis, the mechanisms underlying muscle fatigue are still unclear. Here we do not aim to review the literature or the techniques to investigate NMF as it has been the focus of several recent reviews, including those from our groups [2, 5–7]. Rather, we aim to focus on the pitfalls researchers inevitably face when neuromuscular fatigue is quantified during/after exercise, as to the best of our knowledge it has never been reviewed.

This short review will first reconsider the definition of NMF then discuss some methodological issues that are related to both the type of exercise inducing NMF and the way NMF is assessed. Since electrical stimulation—and in particular transcutaneous supramaximal nerve electrical stimulation—may be painful, it is not always suitable for frail populations. For that reason, we will propose some alternatives and discuss their limitations. We will then present other potential issues with stimulation and recording techniques. Finally, we will provide a summary and some recommendations that could help to better understand NMF.

2 Definition of Neuromuscular Fatigue

There is no clear consensus on the definition of NMF, and this may ultimately affect its quantification and determinants. The most used definition of NMF is probably an exercise-induced reduction in maximal voluntary contraction (MVC) force [7, 8] or the inability to produce a certain expected force or power [8], a definition used in many of our (e.g. [9–11]) and other (e.g. [12, 13]) papers. First, it is important to emphasize that the loss of maximal functional capacities should not be confounded with task failure, as the underlying mechanisms can differ. For instance, the time to task failure of a sustained submaximal isometric contraction performed with the knee extensors correlated with knee extensor EMG activity measured at task failure (i.e. reflecting the intensity of the motor command at this time point), while MVC force loss measured at task failure correlated with the reduction of electrically evoked force, an index of impaired muscle contractile function [9].

Yet even when considering the loss of maximal functional capacities, or transient loss of work [14] or force [15] generating capacities, two main issues can be highlighted. The first one is that it is possible to observe no decrease in MVC force despite there being some level of fatigue. Indeed, low-frequency fatigue [16], also known as prolonged low-frequency force depression [17] (PLFFD), may exist despite MVC force (i.e. high motor units discharge rate) not changing or being fully recovered. For that reason, the definition of MacIntosh and Rassier [18], i.e. 'a response that is less than the expected or anticipated contractile response, for a given stimulation' must be preferred as it encompasses force loss in low-frequency stimulations as well.

In addition, one must keep in mind that when NMF is only measured at exhaustion, one must assume that the motivation of the participants is optimal. This is probably not the case, particularly with clinical populations. We recently proposed a new test with progressive incremental loading and multiple assessments that limits the influence of participant's cooperation and motivation: the quadriceps intermittent fatigue (QIF) test [19]. This test involves measuring MVC force and evoked quadriceps responses via femoral nerve magnetic stimulation before the fatiguing task and after each set of 10 submaximal isometric contractions (5-s on/5-s off; starting at 10% MVC with 10% increments) until task failure. By measuring voluntary and evoked force responses at rest, at iso-task and at exhaustion, this test assesses quadriceps strength, endurance and fatigability in a single session. It has already allowed the testing of various clinical populations such as patients with fibromyalgia syndrome [20] or facioscapulohumeral muscular dystrophy [21].

When considering the 'basic' definition of NMF, it is important to keep in mind that isometric vs. dynamic measurements do not give the same results [22]. For instance, we recently assessed both MVC and maximal power during a 7-s sprint after three different fatiguing cycling exercises: a Wingate test (30 s all out), 10-min at a power output 5% above the second ventilatory threshold (severe intensity) and 90-min at a power output 20% below the first ventilatory threshold (moderate intensity) [23]. The maximal power decreased relatively more than the MVC force after the Wingate test while the opposite was found after the severe- and moderate-intensity tasks. If the question about the most fatiguing exercise among the three tasks described above is raised, the response would be that it is the Wingate if maximal power is chosen as the fatigue index but it is the moderate-intensity task if MVC force is the selected index. As highlighted in our recent meta-analysis, the assessment of power after a fatiguing exercise is also particularly relevant compared with traditional isometric measurements when exploring age-related changes in fatigability [24].

If fatigue is a loss of force-generating capacity, it follows that recovery occurs when this capacity is back to normal. We found that 16 days after a 166-km mountain ultra-marathon with 9500 m of positive and negative elevation, MVC and electrically evoked forces (i.e. peripheral fatigue) have

fully recovered [25]. It is very unlikely that the ability of the runners to perform another prolonged exercise was back to initial values. In addition, we have found that heart rate variability was not back to what was measured before the race (personal data) further suggesting that full recovery had not occurred.

3 Single Joint vs Whole-Body Exercise

Task dependency is a well-known characteristic of fatigue. As the etiology of fatigue is multifactorial, the characteristics of the task determine the underlying mechanism(s) and site(s) associated with fatigue [26, 27]. The main task characteristics are duration and intensity, intermittent vs continuous as well as environmental conditions (temperature, altitude). Two other important characteristics that influence the mechanisms underlying NMF are (1) the mode of contraction (e.g. isometric vs dynamic) and (2) whether the exercise is local vs global (i.e. single-joint vs large muscle mass involved). With this in mind, there are two main paradigms to measure fatigue (Fig. 1): during vs before and after exercise measurements. In the first setup, usually (although not always) in isometric mode and single-joint exercises, the ergometer allows the researchers to both induce and measure fatigue, e.g. sustained isometric contraction for a fixed duration or until task failure. In the second setup, typically cycling or running, there is either a specific ergometer to induce fatigue or the fatiguing task is performed in the field but in both cases, the participant must be transferred to the ergometer where NMF is measured. This delay taken to assess fatigue in whole-body exercises, typically 1–4 min [28–31], is problematic because recovery starts as soon as exercise has stopped. The delay it takes to quantify neuromuscular adjustments, i.e. the duration between exercise termination and testing of the neuromuscular function, can affect both neural [32–35] and contractile [34, 36] properties and might contribute to the lack of consensus on the various models proposed to understand NMF (e.g. [37]). Some authors have tried to minimize this delay and were able to shorten the duration to ~ 10 s (e.g. [38, 39]) but the possibility for the investigators to appropriately secure participants using non-compliant straps can be questioned.

In this context, we recently designed and tested the reliability of an innovative ergometer permitting the start of NMF measurement within 1-s after cycling [40]. The recumbent cycling ergometer permitted the instrumented pedals to be locked instantly in a fixed position similar to what can be performed on an isometric chair. This cycling ergometer allowed for the first time to assess NMF without any recovery period after a dynamic exercise with a large muscle mass (Fig. 1). Despite the recorded EMG parameters being more variable than the measures of force, both parameters showed good to excellent relative reliability. The new cycling ergometer represents overall a major methodological advance and was found to be as reliable as traditional measurements performed with an isometric

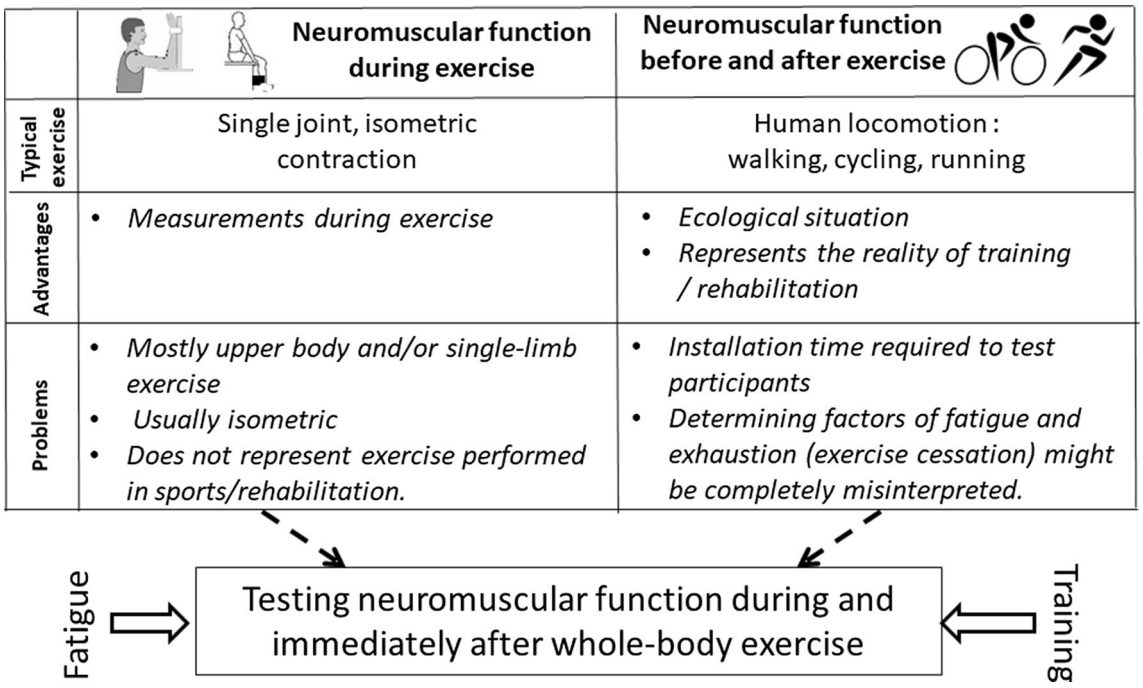

Fig. 1 The two main paradigms to measure fatigue that have led to the proposition of an innovative ergometer [40]

chair [40]. It has already allowed a re-examination of the effects of training on fatigue resistance [41, 42] and to observe evidence of a relationship between the rapid component of VO_2 (τ value) and the extent of NMF, particularly PLFFD [43].

The need for a quick measurement post-fatigue is somehow incompatible with the necessity to perform several contractions to provide a reliable evaluation of MVCs or evoked contractions. The same question exists for the number of contractions per set [44] and the number of sets of contractions to accurately measure voluntary activation with transcranial magnetic stimulation (TMS). This also raises the issue of the appropriate recovery between the different sets or between contractions: if the recovery is too short, fatigue may accumulate due to repeated contractions and if it is too long, some recovery may occur. The question may be even more critical for the number of stimulations needed to accurately measure EMG responses to TMS or to cervicomedullary/thoracic stimulations, i.e. motor-evoked potentials (MEPs), cervicomedullary/thoracic motor-evoked potential (CMEP, TMEP). A possibility that must be investigated in future methodological studies is the increase in sample size to offset the reduction in reliability when only a couple of contractions and/or stimulations are possible due to quick recovery process.

4 Surrogate of Nerve Electrical Stimulation

The techniques used to evaluate neuromuscular fatigue might be painful and, therefore, might lead to biased measurements (see for instance [45] vs. [46]). Alternatives using muscle stimulation [47–50] or magnetic nerve stimulation [51, 52] have been proposed to limit discomfort and evaluate exercise-induced neuromuscular adjustments, yet some limitations exist for these techniques and this must be discussed [53, 54]. This is particularly relevant for clinical studies involving patients or elderly participants, as it is known that fatigue is the main complaint in many chronic diseases [2, 3, 55] and aging [56].

The gold standard to assess neuromuscular properties consists of the transcutaneous supramaximal stimulation of a motor nerve (often a mixed nerve, as is the case for the femoral nerve or tibial nerve for the evaluation of the knee extensors and plantar flexors, respectively). However, this stimulation can be perceived as painful for some (frail) participants, and this is especially the case when more than one stimulus is used. For instance, paired stimulation is commonly used when imposed during a maximal voluntary contraction to increase the signal to noise ratio and in turn provide a better estimation of the superimposed force using the twitch interpolation technique [57]. Supramaximal stimulation of a motor nerve might be painful. It has been suggested that expectation of a noxious stimulus may induce significant decrements in force and maximal voluntary activation measured with the twitch interpolation technique and, therefore, might lead to biased measurements of actual MVC force and maximal voluntary activation [45]. However, it seems that this may happen only if participants have no previous experience with stimulation [45, 46]. Indeed, we have demonstrated that in familiarized participants, knowing a stimulus (not only supramaximal nerve stimulation but also muscle stimulation and TMS) will occur does not impair MVC force or maximal voluntary activation despite differences in discomfort associated with the stimulus modalities [46]. Although the level of discomfort does necessarily jeopardize the measurement of MVC force and maximal voluntary activation, methods have been proposed in the last few years to reduce pain during stimulation (either superimposed or on relaxed muscles) while providing a valid quantification.

Muscle stimulation is one of these methods although it has been suggested that it may have some limitations. Using large electrodes positioned proximally on the knee extensors, it is possible to overcome the usual limitations of (1) incomplete recruitment associated with muscle stimulation (e.g. [58, 59]) and (2) inability to obtain the M wave due to the proximity of stimulating and recording EMG electrodes. Maximal twitch and M-wave amplitudes (vastus lateralis and vastus medialis muscles) were similar using muscle stimulation compared to maximal femoral nerve stimulation but the level of discomfort associated with the stimulation was consistently lower [60]. Similarly, we have previously shown that the use of muscle stimulation gives a comparable estimate of muscle inactivation (i.e. superimposed doublet) during contractions $\geq 60\%$ maximal voluntary force in the knee extensors, and this was again associated with a lower discomfort score [59]. Using muscle stimulation, Bampouras et al. [47] showed that an intensity eliciting 50% of the maximal twitch amplitude gives comparable results of muscle activation (central activation ratio) as compared to higher stimulation intensities. Thus, it appears possible to obtain valid measurements of neural and contractile function while reducing the discomfort associated with the technique, which is definitely helpful for testing in frail populations. We have also shown that submaximal tetanic muscle stimulation was valid to assess PLFFD in the knee extensors after eccentric exercise [48]. Similarly, the use of paired stimulation rather than supramaximal electrical tetanic stimulation has been validated for the evaluation of PLFFD in the knee extensors [52]. However, the use of paired stimuli as a surrogate of contractile adaptations to exercise should be reassessed by other groups as a recent report suggests that the use of a doublet may actually underestimate the level of PLFFD in the dorsiflexor muscles [61] and it has not been assessed for high-frequency fatigue. Thus, if discomfort is an issue for the participants, knee extensor neuromuscular function

may be assessed via large electrodes positioned proximally instead of femoral nerve stimulation. If supramaximal muscle stimulation is still an issue, submaximal intensities may be used for the assessment of muscle activation level or PLFFD.

Interestingly, both magnetic and electrical paired stimuli delivered over the nerve provided a valid assessment of contractility in the resting and fatigued states in both the knee extensors [52] and plantar flexors [51], the advantage of the magnetic nerve stimulation being that it is usually perceived as less uncomfortable [51, 62]. Provided an appropriate coil is used [54], the use of magnetic nerve stimulation offers interesting perspectives for the evaluation of frail individuals but the size of the coil used for stimulation restrains the position to be used for testing the participant [51, 52]. One major limitation, however, is that it may not be suitable for neurogenic conditions such as Charcot–Marie–Tooth disease [63] or for overweight participants given the maximal stimulation output of the current magnetic stimulators [53]. Indeed, the stimulators may not have sufficient field strength to excite all motor axons in people with a large subcutaneous fat layer. Apart from the knee extensors [52] and plantar flexors [51], the use of magnetic stimulation to assess muscle contractile properties has also been validated (i.e. showing similar force levels compared with electrical nerve stimulation) in small muscles such as adductor pollicis [64]. Nevertheless, as the non-focal stimulation induced by the relatively large size of the coil may potentially activate antagonist nerves or muscles (and thus underestimate the evoked force), any use in other conditions or muscle groups should first be validated against motor nerve stimulation.

5 Other Potential Issues with Stimulation/Recording Techniques and Data Analysis

Some other potential problems can arise from the techniques themselves. For instance, we have recently shown that changes in the parameters obtained with TMS are stimulation intensity dependent. MEP, an index of corticospinal excitability, can be determined using a given percentage of resting or active motor threshold, or using a MEP stimulus–response curve. Those techniques clearly showed different outcomes [65], which make the comparison between studies in the literature difficult. Even when considering only the stimulus–response curve method, one can stimulate at optimal TMS intensity (defined as the lowest stimulus intensity eliciting maximal MEP amplitudes, i.e. when all upper motoneurons are recruited) or at submaximal intensity (corresponding to the rising part of the MEP stimulus–response curve, i.e. when only a portion of the upper motoneurons are recruited). One may speculate that MEP evoked at submaximal intensity is more likely to increase when a submaximal intensity is used because there is more room for a raise. However, this is likely a simplistic approach. It has been argued that submaximal intensity (i.e. weak stimulus) tests the responsiveness of the small motoneurones [66]. We showed that a mountain ultra-marathon induced an increase in MEP amplitude when measured at optimal TMS intensity but MEP was not modified when suboptimal TMS intensity (equivalent to 60% of the optimal intensity) was used [67]. Similarly, the stimulation intensity gave different outcomes for the silent period (SP), an index of corticospinal inhibition, i.e. no change in SP duration at optimal intensity but an increase at submaximal intensity [67]. In another study, we demonstrated that TMS intensity affects exercise-induced changes in MEP amplitude/M wave amplitude (only when measured at absolute force level), silent period, and voluntary activation after a fatiguing exercise consisting of repetitive submaximal isometric knee extensions at 40% MVC force until task failure [68]. We suggest that when TMS is used to assess voluntary activation, optimal TMS intensity is preferred. We discourage using supramaximal intensity, as for nerve stimulation, because of the risk of coactivation [65]. On the contrary, when the question relates to a fatigue specific to the type of motoneuron activated it is recommended that both maximal and submaximal intensities are used. However, researchers will have to choose between delivering either (1) both types of stimulations in the same sets or (2) delivering maximal stimulations in one session and submaximal stimulations in a separate session. Both options have advantages and disadvantages and the decision must be made based on the scientific question and experimental design (e.g. type of fatiguing exercise).

Data analysis may also be problematic, as is the case for M-wave amplitude [69] or voluntary activation level measurement [70–72]. A reduction in M-wave peak-to-peak amplitude has been very widely used as an index of impaired sarcolemmal excitability over the last 40 years. Recent findings from our laboratory indicate that this may lead to inappropriate conclusions, as the electrical formation of the first and second phases of an M wave is intrinsically different (for a review, see [73]). Using different exercise paradigms (e.g. [69, 74]), we recently suggested that a separate analysis of the first and second phases of the M wave should be conducted and that the increase (and not decrease) of the first phase of the M wave reflects impaired muscle membrane excitability. This challenging hypothesis might have significant implications for our understanding of NMF.

In the twitch interpolation technique [75], any increase in force due to the superimposed electrical stimulation over a MVC is considered as motor units not discharging maximally or not being spatially recruited during the (maximal) voluntary contraction [76]. The twitch interpolation technique has been debated in the literature [77, 78]. Its interpretation has also been challenged 10 years ago when increased

superimposed forces were observed in fatigued mouse intact single fibers [72]. These findings could obviously not be explained by impaired neural drive in this preparation but were explained by the fact that muscle fibers entered a steeper part of the force–intracellular Ca^{2+} relationship during fatigue. This observation suggests that the twitch interpolation technique could overestimate the extent of central fatigue. Although these results were not confirmed using tetanic stimulation of the ulnar nerve in humans [70], we showed in another study that peripheral fatigue factors can affect the superimposed twitch force [71].

6 Conclusions and Recommendations for Future Studies

This review, summarized in Fig. 2, shows that current practices in the assessment of NMF can be problematic in many ways, although alternatives have been proposed to optimize its quantification. From this figure, it can be appreciated that there is probably no "gold standard" to evaluate exercise-induced acute changes in both neural and contractile properties, i.e. the assessment depends on the scientific question. However, there are definitely recommendations to be proposed to improve the characterization of NMF:

1. Familiarize the participant during a separate session (particularly if electrical/magnetic stimulation is used)
2. Combine various approaches to characterize NMF (e.g. loss in isometric MVC force vs maximal power/maximal velocity, time trial, time to task failure, QIF test). Single-joint sustained maximal isometric contraction is a valid model of fatiguing exercise but it does not represent the reality of most activities.
3. Consider supramaximal electrical stimulation over the muscle via large electrodes located proximally to characterize neural and contractile properties of the knee extensors if the discomfort associated with supramaximal nerve stimulation is an issue. Submaximal intensities may even be used for the assessment of muscle activation level or PLFFD.
4. Consider magnetic nerve stimulation of the motor nerve for the plantar flexors and knee extensors in frail populations, although the size of the coil constrains the position in which the participant has to be tested. Validation against supramaximal electrical nerve stimulation is

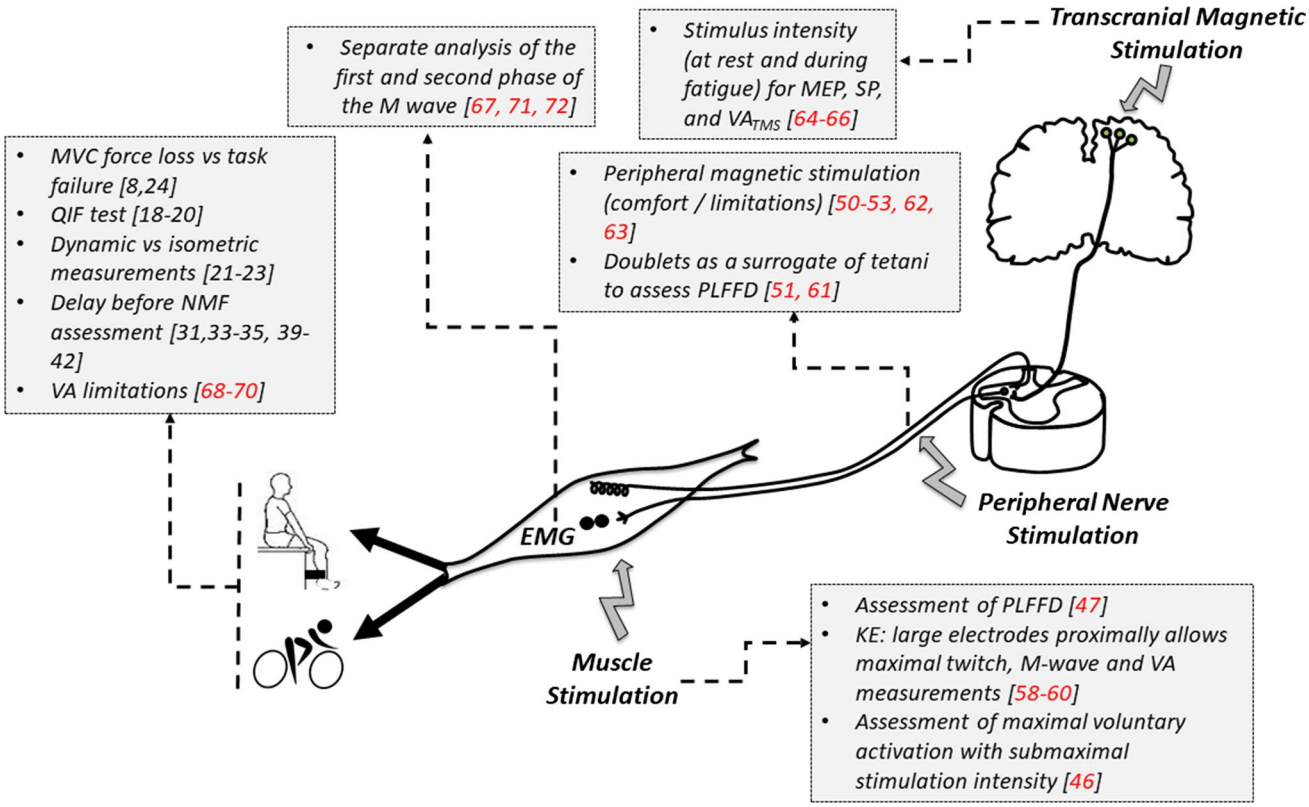

Fig. 2 Summary of some of the issues/alternatives in the current practices to evaluate neuromuscular fatigue. *EMG* electromyography, *MEP* motor evoked potential, *MVC* maximal voluntary contraction, *NMF* neuromuscular fatigue, *PLFFD* prolonged low-frequency force depression, *QIF* quadriceps intermittent fatigue, *SP* silent period, *TMS* transcranial magnetic stimulation, *VA* maximal voluntary activation

required before implementation in any other condition. Confirmation of supramaximality and measurement of antagonist activity should be performed in all participants.

5. Consider a separate analysis of the first and second phases of the M wave, as the amplitude of the first phase may be more sensitive to changes in sarcolemmal excitability than the usual peak-to-peak amplitude assessment.
6. As the twitch interpolation technique may lead to overestimation of the extent of central fatigue, it is recommended to combine this measurement with other indices of neural adaptations based on surface EMG parameters (e.g. maximal EMG activity normalized by M-wave amplitude/MEP/volitional (V) wave/Hoffmann reflex, depending on the research question).
7. TMS intensity must be chosen with consideration of the research question, i.e. optimal intensity when the goal is to assess voluntary activation at the cortical level vs submaximal and maximal intensity if the goal is to investigate the behaviors of different motoneuron pools.

These recommendations might be considered as 'best practice' to optimize the detection, quantification and interpretation of NMF in future works. Finally, while NMF assessment is important to investigate the (patho-)physiology of human beings performing an exercise, it must be kept in mind that it can only partially explain performance so more comprehensive models must be used (e.g. [79]).

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Compliance with Ethical Standards

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